



A Level • Cambridge (CIE) • Chemistry

28 mins

15 questions

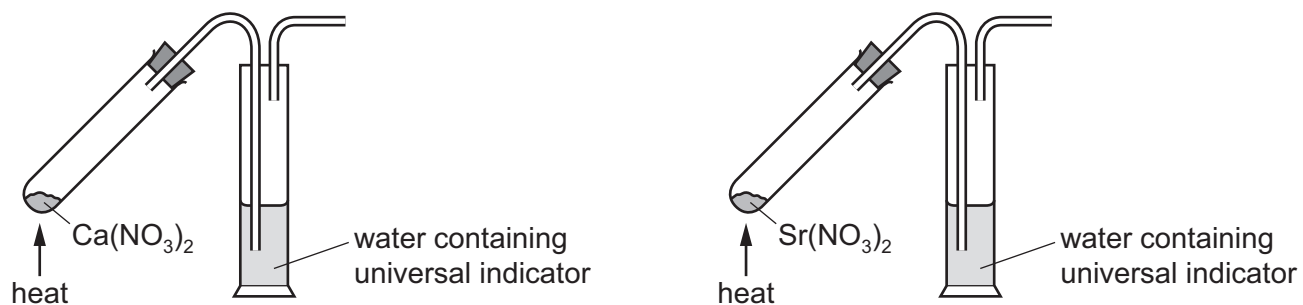
Multiple Choice Practice

Topical Practice MCQ

10. Group 2

Carefully selected past paper questions by topic.

- 22 The diagram shows the process of adding calcium nitrate and strontium nitrate to separate boiling tubes and heating them. Identical conditions are used.



As the reactions proceed, the water containing universal indicator changes colour.

Which row describes the colour change and identifies the nitrate that causes the quickest colour change?

	colour change of universal indicator	nitrate that causes the quickest colour change
A	green to blue	$\text{Ca}(\text{NO}_3)_2$
B	green to blue	$\text{Sr}(\text{NO}_3)_2$
C	green to red	$\text{Ca}(\text{NO}_3)_2$
D	green to red	$\text{Sr}(\text{NO}_3)_2$

- 14** Q is a mixture of a Group 2 oxide and a Group 2 sulfate. Q contains equal amounts of the two compounds.

Q is shaken with water and the resulting mixture filtered; a solid residue is obtained. There is no reaction when the solid residue is shaken with $\text{HCl}(\text{aq})$. Shaking the filtrate with $\text{H}_2\text{SO}_4(\text{aq})$ produces a white precipitate.

What could be Q?

- A** $\text{BaO} + \text{BaSO}_4$
- B** $\text{BaO} + \text{MgSO}_4$
- C** $\text{MgO} + \text{BaSO}_4$
- D** $\text{MgO} + \text{MgSO}_4$

Question 3

9701_w21_qp_11 • Q15

- 15** Anhydrous magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, decomposes when heated, giving a white solid and a mixture of two gases, X and Y.

Y is oxygen.

What is the ratio $\frac{\text{mass of X released}}{\text{mass of Y released}}$?

A $\frac{1}{0.174}$

B $\frac{1}{0.267}$

C $\frac{1}{0.348}$

D $\frac{1}{3.43}$

Question 4

9701_s19_qp_12 • Q15

- 15** How many of the solutions shown, when added to separate portions of magnesium sulfate solution, produce a white precipitate?

HCl(aq) $\text{NH}_3(\text{aq})$ $(\text{NH}_4)_2\text{CO}_3(\text{aq})$ $\text{Ba}(\text{NO}_3)_2(\text{aq})$

A 0

B 1

C 2

D 3

- 15 A solution contains both $\text{Mg}^{2+}(\text{aq})$ and $\text{Sr}^{2+}(\text{aq})$ at the same concentration.

The solution is divided into two equal portions. Aqueous sodium hydroxide is added dropwise to one portion. Dilute sulfuric acid is added dropwise to the other portion.

Which row is correct?

	precipitate seen first when $\text{NaOH}(\text{aq})$ is added	precipitate seen first when $\text{H}_2\text{SO}_4(\text{aq})$ is added
A	magnesium hydroxide	magnesium sulfate
B	magnesium hydroxide	strontium sulfate
C	strontium hydroxide	magnesium sulfate
D	strontium hydroxide	strontium sulfate

- 18** When 3.00g of an anhydrous nitrate of a Group 2 metal is decomposed, 1.53g of gas is produced.

What is the nitrate compound?

- A** beryllium nitrate
- B** calcium nitrate
- C** magnesium nitrate
- D** strontium nitrate

- 14** A 0.005 mol sample of anhydrous calcium carbonate was completely thermally decomposed to give 100 cm³ of gas measured at a certain temperature and pressure.

In a separate experiment carried out at the same temperature and pressure, a 0.005 mol sample of anhydrous calcium nitrate was completely thermally decomposed. The volume of gaseous products was measured.

What total volume of gaseous products was produced from the calcium nitrate?

- A** 50 cm³ **B** 100 cm³ **C** 200 cm³ **D** 250 cm³

16 A solid, **T**, was placed in an excess of the liquid **U**.

A colourless gas was given off and a white precipitate was seen. The precipitate was not **T**.

What could be the identities of **T** and **U**?

	T	U
A	BaCO ₃	H ₂ O
B	Ca	dilute H ₂ SO ₄
C	Mg	dilute H ₂ SO ₄
D	SrCO ₃	dilute HCl

15 When calcium is burnt in oxygen, what colour is the flame?

- A** green
- B** red
- C** white
- D** yellow

12 When barium is burnt in oxygen, what colour is the flame?

- A** green
- B** orange
- C** red
- D** white

Question 11

9701_s13_qp_12 • Q14

- 14 In which row of the table are all statements comparing the compounds of calcium and barium correct?

	solubility of calcium hydroxide	solubility of barium hydroxide	thermal stability of calcium carbonate	thermal stability of barium carbonate
A	higher	lower	higher	lower
B	higher	lower	lower	higher
C	lower	higher	higher	lower
D	lower	higher	lower	higher

12 Use of the Data Booklet is relevant to this question.

Anhydrous magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$, will decompose when heated, giving a white solid and a mixture of two gases **X** and **Y**.

Y is oxygen.

What is the ratio $\frac{\text{mass of X released}}{\text{mass of Y released}}$?

A $\frac{1}{0.174}$

B $\frac{1}{0.267}$

C $\frac{1}{0.348}$

D $\frac{1}{3.43}$

- 13** A metal, **X**, reacts with water to produce a colourless solution which gives a white precipitate when mixed with aqueous sulfuric acid.

What is metal **X**?

- A** barium
- B** magnesium
- C** potassium
- D** sodium

Question 14

9701_w12_qp_11 • Q14

- 14 What are the trends in the stated properties as Group II is descended from magnesium to barium?

	decomposition temperature of the carbonate	first ionisation energy
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 12 Equimolar quantities of magnesium carbonate and strontium carbonate are separately heated to bring about complete thermal decomposition. The minimum temperature for this to occur is called T_d .

The cold residues are separately added to equal volumes of water and the change in pH is measured. The change in pH is called ΔpH .

Which metal has the higher value of T_d , and the greater value of ΔpH ?

	T_d	ΔpH
A	Mg	Mg
B	Mg	Sr
C	Sr	Mg
D	Sr	Sr

Answer Key

No.	Paper	Topic	Answer
1	9701_m24_qp_12	10. Group 2	C
2	9701_s21_qp_11	10. Group 2	A
3	9701_w21_qp_11	10. Group 2	A
4	9701_s19_qp_12	10. Group 2	D
5	9701_m18_qp_12	10. Group 2	B
6	9701_s18_qp_11	10. Group 2	D
7	9701_s16_qp_12	10. Group 2	D
8	9701_s16_qp_12	10. Group 2	B
9	9701_s14_qp_11	10. Group 2	B
10	9701_s14_qp_12	10. Group 2	A
11	9701_s13_qp_12	10. Group 2	D
12	9701_s13_qp_13	10. Group 2	A
13	9701_w13_qp_11	10. Group 2	A
14	9701_w12_qp_11	10. Group 2	C
15	9701_w10_qp_12	10. Group 2	D